

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

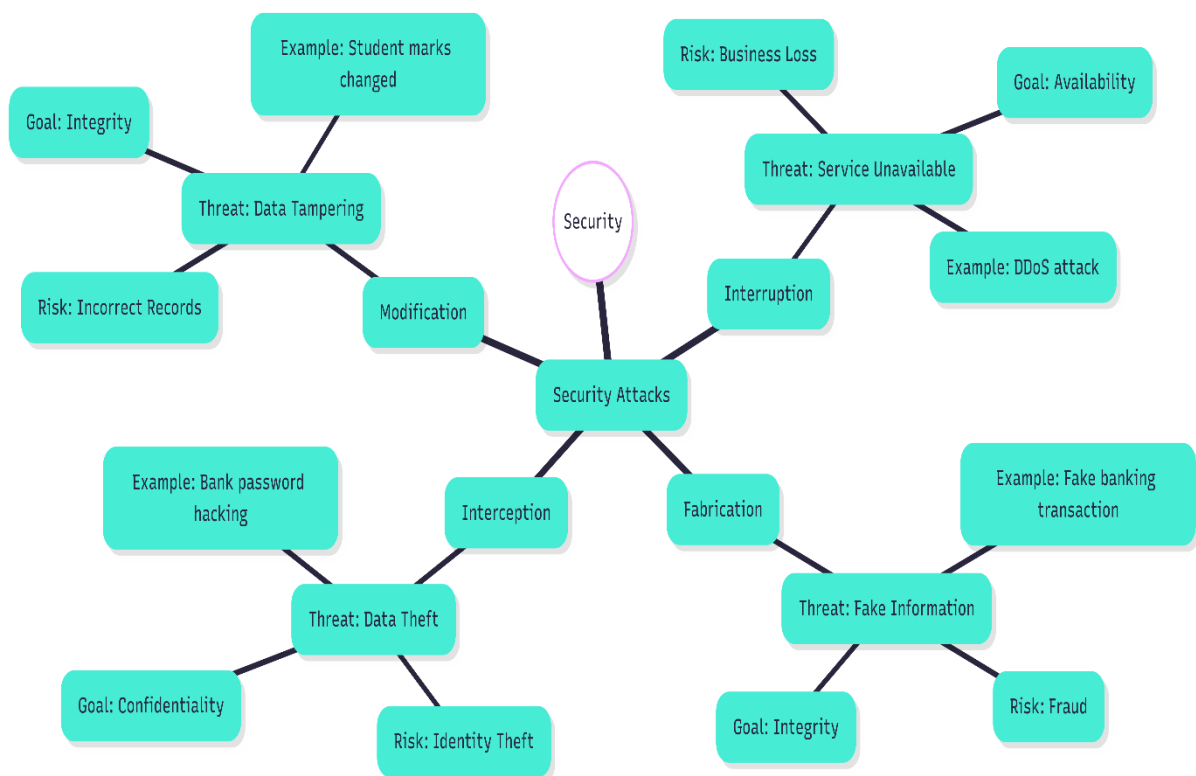
Signature of the student: D. Gnanadeepthi

Annexure A

CONCEPT MAPPING

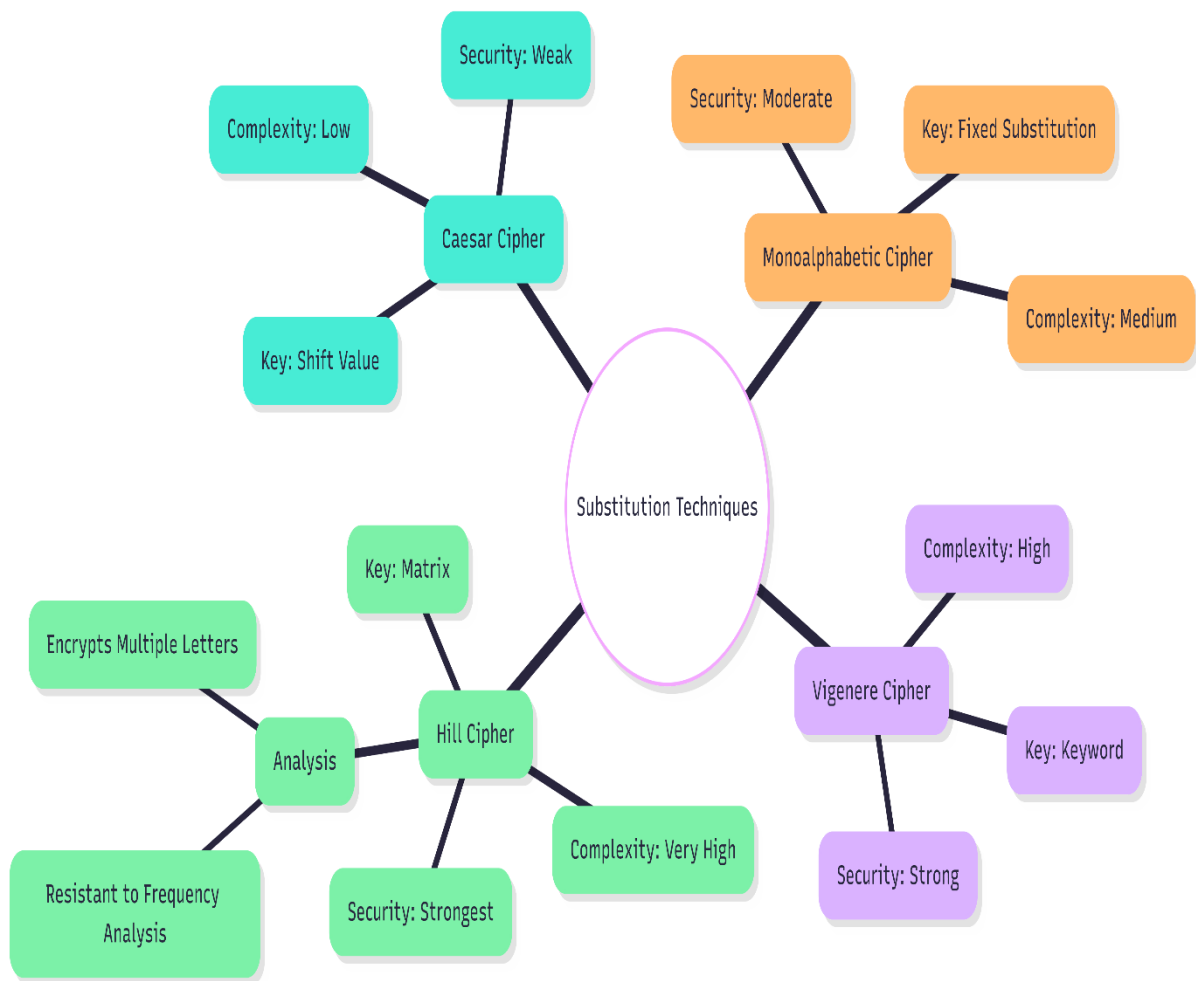
1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link



Attack	Threat	Risk	Security Goal	Real-world Example
Interception	Data theft	Identity theft	Confidentiality	Hacker stealing bank passwords
Modification	Data tampering	Incorrect records	Integrity	Changing student marks
Interruption	Denial of Service	Website unavailable	Availability	DDoS attack on Amazon
Fabrication	Fake messages	Fraud	Integrity	Fake banking transaction

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

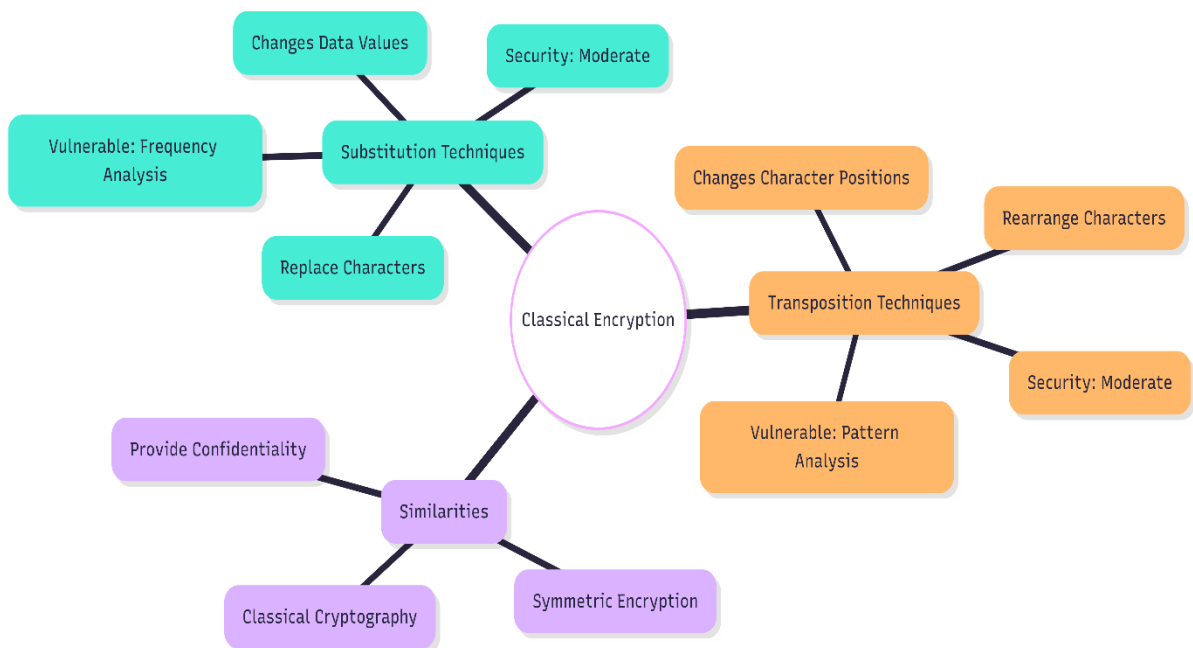


Cipher	Key Usage	Complexity	Security
Caesar	Shift key	Low	Very Weak
Monoalphabetic	One substitution alphabet	Medium	Weak
Vigenère	Keyword	High	Strong
Hill	Matrix key	Very High	Strongest

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why



Feature	Substitution	Transposition
Changes	Characters	Positions
Data Structure	Values changed	Order changed
Security	Moderate	Moderate
Vulnerable To	Frequency analysis	Pattern analysis

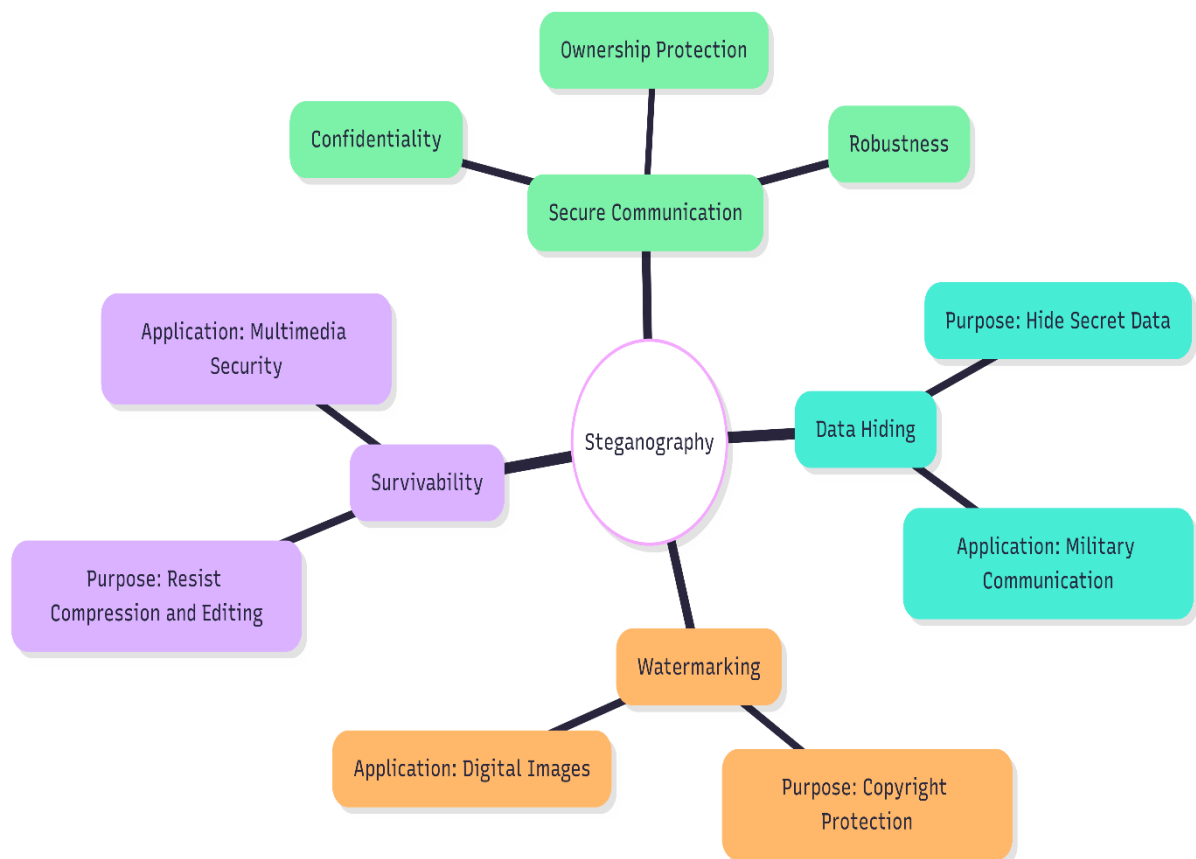
Analysis

More Vulnerable: Substitution Cipher

Reason: Letter frequencies remain unchanged, making frequency analysis effective

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each



Concept	Purpose	Real-world Application
Data Hiding	Hide secret data	Secret military communication
Watermarking	Copyright protection	Digital image copyright
Survivability	Resist modifications	Secure multimedia transmission

Analysis

- Data hiding protects confidentiality.
- Watermarking proves ownership.
- Survivability ensures hidden data remains after compression or editing.

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify



Mathematical Connections

- Extended Euclid → Finds Modular Inverse
- Euler Phi Function → Used in Euler's Theorem
- Euler's Theorem → Basis of RSA
- Fermat's Little Theorem → Special case of Euler's Theorem

Role in Encryption:

Concept	Role
Extended Euclid	Computes decryption key
Modular Inverse	Finds private key
Fermat's Theorem	Modular exponentiation
Euler Phi Function	Generates RSA keys
Euler's Theorem	Encryption and decryption correctness

Analysis

Most Critical: Euler's Phi Function ($\phi(n)$)

Justification:

- Essential for RSA key generation.
- Used to compute public and private keys.
- Without $\phi(n)$, RSA encryption and decryption cannot be performed.

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginning
Understan ding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/tec hnologies	Good understanding with proper integration of most concepts	Basic understand ing; limited or partial integration	Poor understan ding; unable to integrat e concepts
Experimen t al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implement ation with errors or inefficienci es	Incorrect or incomplet e implement ation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct executio n	Limited or inconsisten t use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable inter pretation	Basic analysis with limite d insights	Poor or incorrect analysis of results

Document ation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documenta tion with missing details	Poor or incomplet e document ation
-------------------	----	---	--	---	--